

4.3.51

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Question

The equation of a line is $3x - 4y + 10 = 0$. Find its slope and the x and y intercepts.

Solution

Let $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$. Then

$$\begin{pmatrix} 3 & -4 \end{pmatrix} \mathbf{x} = -10 \quad (1)$$

Comparing with $\mathbf{n}^T \mathbf{x} = c$, we identify

$$\mathbf{n} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}, \quad c = -10. \quad (2)$$

A direction vector orthogonal to $\mathbf{n}$ is

$$\mathbf{m} = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \quad (\mathbf{n}^T \mathbf{m} = 3 \cdot 4 + (-4) \cdot 3 = 0). \quad (3)$$

Solution

Hence the slope is

$$\text{slope} = \frac{m_y}{m_x} = \frac{3}{4}. \quad (4)$$

x-intercept: points on the x-axis have $\mathbf{x} = \begin{pmatrix} x \\ 0 \end{pmatrix}$. Substituting in (1),

$$\begin{pmatrix} 3 & -4 \end{pmatrix} \begin{pmatrix} x \\ 0 \end{pmatrix} = -10 \Rightarrow 3x = -10 \Rightarrow x = -\frac{10}{3}. \quad (5)$$

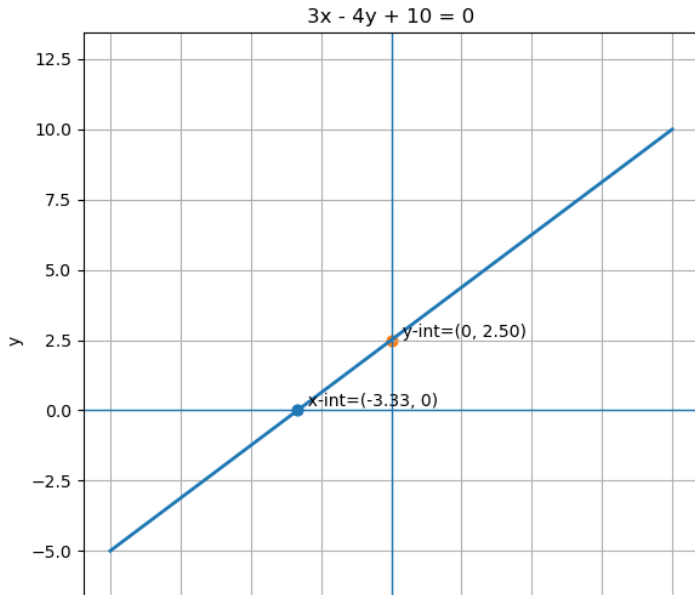
$$\text{x-intercept} = \begin{pmatrix} -\frac{10}{3} \\ 0 \end{pmatrix}. \quad (6)$$

y-intercept: points on the y -axis have $\mathbf{x} = \begin{pmatrix} 0 \\ y \end{pmatrix}$. Substituting in (1),

$$\begin{pmatrix} 3 & -4 \end{pmatrix} \begin{pmatrix} 0 \\ y \end{pmatrix} = -10 \Rightarrow -4y = -10 \Rightarrow y = \frac{5}{2}. \quad (7)$$

$$\text{y-intercept} = \begin{pmatrix} 0 \\ \frac{5}{2} \end{pmatrix}. \quad (8)$$

Graph



```
#include <math.h>

// ax + by + c = 0  slope = -a/b (if b != 0)
// x-intercept = (-c/a, 0) if a != 0
// y-intercept = (0, -c/b) if b != 0
int line_params(double a, double b, double c,
               double* slope, int* slope_defined,
               double* xint, int* x_defined,
               double* yint, int* y_defined)
{
    if (b != 0.0) { *slope = -a/b; *slope_defined = 1; }
    else { *slope = NAN; *slope_defined = 0; }

    if (a != 0.0) { *xint = -c/a; *x_defined = 1; }
    else { *xint = NAN; *x_defined = 0; }
```

```
if (b != 0.0) { *yint = -c/b; *y_defined = 1; }
else { *yint = NAN; *y_defined = 0; }

return 0;
}

// Convenience: for  $3x - 4y + 10 = 0$ 
int example_3m4p10(double* slope, int* slope_defined,
                  double* xint, int* x_defined,
                  double* yint, int* y_defined)
{
    return line_params(3.0, -4.0, 10.0,
                      slope, slope_defined,
                      xint, x_defined,
                      yint, y_defined);
}
```


Python Code

```
import numpy as np
import matplotlib.pyplot as plt

# Line:  $3x - 4y + 10 = 0$   $y = (3/4)x + 10/4$ 
a, b, c = 3.0, -4.0, 10.0

# Compute intercepts (if defined)
x_int = -c/a if a != 0 else None # (-10/3, 0)
y_int = -c/b if b != 0 else None # (0, 2.5)

# Choose a plotting window around intercepts
xs = np.linspace(-10, 10, 400)
if b != 0:
    ys = (-a*xs - c) / b
else:
    # vertical line  $x = -c/a$ 
    xs = np.full_like(xs, -c/a)
    ys = np.linspace(-10, 10, 400)
```

```
# Plot the line
plt.plot(xs, ys, linewidth=2)
plt.title(3x - 4y + 10 = 0)

# Axes crosshairs
plt.axhline(0, linewidth=1)
plt.axvline(0, linewidth=1)

# Mark intercepts if present
if x_int is not None:
    plt.scatter([x_int], [0], s=40)
    plt.text(x_int, 0, f x-int=({x_int:.2f}, 0), va=bottom)
```

```
if y_int is not None:
    plt.scatter([0], [y_int], s=40)
    plt.text(0, y_int, f y-int=(0, {y_int:.2f}), va=bottom)

plt.xlabel(x)
plt.ylabel(y)
plt.axis(equal)
plt.grid(True)
plt.tight_layout()
```

```
import ctypes
from ctypes import c_double, c_int, byref

# Adjust name for macOS if needed: ./liblinegeom.dylib
lib = ctypes.CDLL('./liblinegeom.so')

lib.line_params.argtypes = [c_double, c_double, c_double,
                             ctypes.POINTER(c_double), ctypes.
                             POINTER(c_int),
                             ctypes.POINTER(c_double), ctypes.
                             POINTER(c_int),
                             ctypes.POINTER(c_double), ctypes.
                             POINTER(c_int)]

lib.line_params.restype = c_int
```

```

lib.example_3m4p10.argtypes = [ctypes.POINTER(c_double), ctypes.
    POINTER(c_int),
                                ctypes.POINTER(c_double), ctypes.
                                POINTER(c_int),
                                ctypes.POINTER(c_double), ctypes.
                                POINTER(c_int)]

lib.example_3m4p10.restype = c_int
def run_example():
    slope = c_double(); slope_def = c_int()
    xint = c_double(); x_def = c_int()
    yint = c_double(); y_def = c_int()

    lib.example_3m4p10(byref(slope), byref(slope_def),
                       byref(xint), byref(x_def),
                       byref(yint), byref(y_def))

```

```
print(Line: 3x - 4y + 10 = 0)
if slope_def.value: print(fSlope m = {slope.value:.6g})
if x_def.value: print(fx-intercept = ({xint.value:.6g}, 0))
if y_def.value: print(fy-intercept = (0, {yint.value:.6g}))

def run_general(a, b, c):
    slope = c_double(); slope_def = c_int()
    xint = c_double(); x_def = c_int()
    yint = c_double(); y_def = c_int()

    lib.line_params(a, b, c,
                    byref(slope), byref(slope_def),
                    byref(xint), byref(x_def),
                    byref(yint), byref(y_def))
```

```
print(fLine: {a}x + {b}y + {c} = 0)
    if slope_def.value: print(fSlope m = {slope.value:.6g})
    else: print(Slope undefined (vertical line))
    if x_def.value: print(fx-intercept = ({xint.value:.6g}, 0))
    else: print(No finite x-intercept (a = 0))
    if y_def.value: print(fy-intercept = (0, {yint.value:.6g}))
    else: print(No finite y-intercept (b = 0))

if __name__ == '__main__':
    run_example()
    # run_general(3.0, -4.0, 10.0)
```